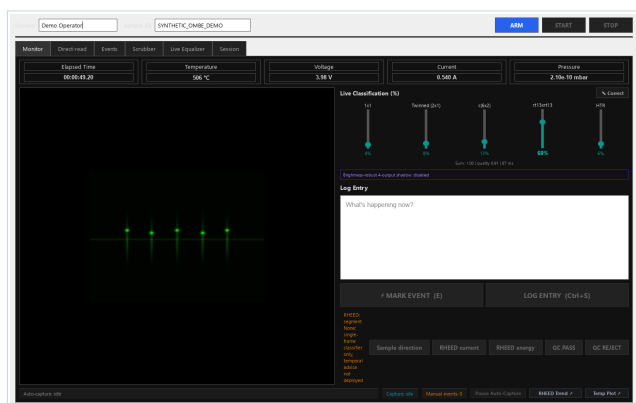


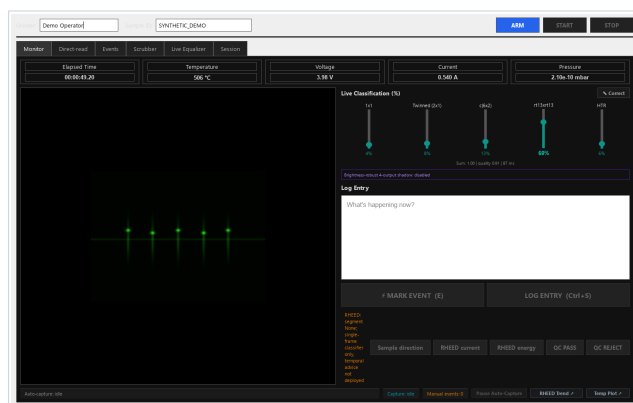
O-MBE + Ch-MBE GUI and RHEED Post-processing Labeling

English Operator Manual



O-MBE Growth Monitor

Generated demo inputs; no hardware.



Ch-MBE Growth Monitor

Generated demo inputs; no hardware.

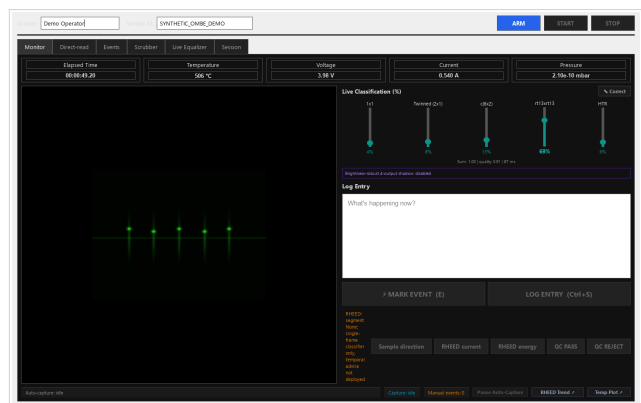
The post-processing labeler reads archived data only. It does not connect to or control cameras, pyrometers, power supplies, or any other instrument.

Quick entry	Double-click
O-MBE live GUI	Start O-MBE Growth Monitor.cmd
Ch-MBE live GUI	Start Ch-MBE Growth Monitor.cmd
RHEED offline labeler	Start RHEED Post-processing Labeler.cmd
Install or refresh desktop shortcuts	Install AI4MBE Desktop Shortcuts.cmd

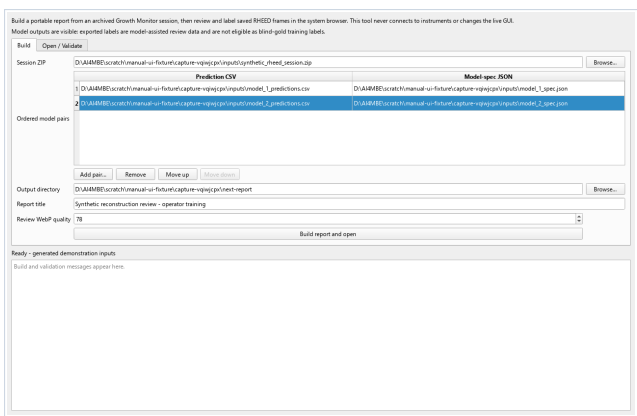
1 Understand the two independent paths

In brief. The live GUI acquires data and writes session records. The offline labeler reads completed archives later and never controls instruments.

The live GUI performs acquisition and display. The offline labeler works only on completed archived sessions. Never interpret an offline result as a real-time control instruction.



1. Live acquisition
The chamber-selected Growth Monitor writes the archive.



2. Offline build and validation
The labeler combines the ZIP with predictions and model specifications.

The live GUI creates session records. The labeler reads those records later, produces a browser report, and exports validated JSON. It never sends commands back to the instruments.

What this manual covers

Chapter	Topic	Outcome
2	Chamber-specific live startup	Select the correct launcher and verify its exact title
3	Separate O-MBE and Ch-MBE defaults	Record approved values without cross-copying
4	Labeler desktop application	Select inputs, build, open, and validate
5-6	Inputs and safe report build	Produce a complete local report directory
7-8	Segment labeling and display controls	Create traceable labels on actual saved frames
9	Export, import, and validation	Produce canonical JSON and validate fail-closed
10-12	Safety, troubleshooting, and version checks	Handoff and reproduce the work safely

Surface-reconstruction labels, acquisition-quality QC, and FeSe film quality are three different concepts. The current classifier concerns the bare STO surface before growth.

Operating principle

- The selected O-MBE or Ch-MBE GUI reads live instrument sources and writes chamber-identified session records under the operator's control.
- Post-processing combines an archived session with one or more already-generated prediction tables.
- Human labels are stored as frame-bound temporal segments with provenance.
- Model outputs remain visible, so these annotations are assisted review, not blind-gold labels.

2 Install shortcuts and start the correct chamber GUI

In brief. Choose exactly one chamber launcher, verify its exact window title, and check live status before ARM. The launchers isolate chamber identity, but they do not authorize operation or setpoint changes.

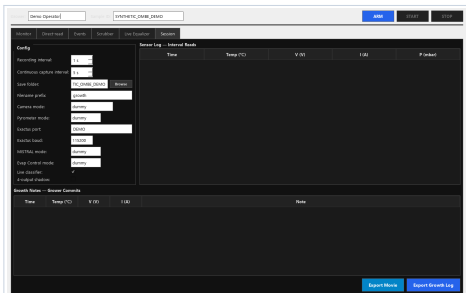
Shared first setup or after moving the checkout

1. From the GUI repository root, double-click `Install AI4MBE Desktop Shortcuts.cmd`.
2. Confirm that the desktop contains **O-MBE Growth Monitor**, **Ch-MBE Growth Monitor**, and **RHEED Post-processing Labeler**.
3. If Windows displays an unexpected security warning, stop and ask the maintainer to verify the file source, branch, and commit. Do not bypass it.

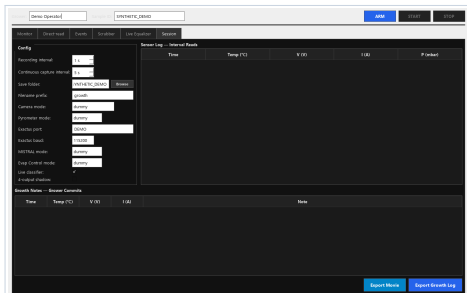
Select exactly one chamber branch

Physical chamber	Required launcher	Exact window title
O-MBE	Start O-MBE Growth Monitor.cmd	Oxide MBE Growth Monitor
Ch-MBE	Start Ch-MBE Growth Monitor.cmd	Chalcogenide MBE Growth Monitor

A different title or a launcher that does not match the physical chamber is a stop condition. Close normally and restart with the correct launcher; changing a GUI field is not a substitute.



O-MBE Session tab. Generated demo inputs; not operating defaults.



Ch-MBE Session tab. Generated demo inputs; not operating defaults.

Shared ARM and exit boundary

1. Confirm the title-selected chamber, advancing RHEED sequence, valid temperature state, intended log directory, and every selected interface against the matching owner-approved register in Chapter 3. A pending field is not a default.
2. ARM or start only when the authorized operator and applicable SOP permit it. Double-click startup authorizes no instrument operation or setpoint change.
3. Close normally and wait for the window to disappear. Do not force-kill Python while logs are being written.

Each launcher forces and preflights its chamber profile and owns a chamber-specific single-instance mutex. Logs below `%LOCALAPPDATA%\AI4MBE\LauncherLogs` record Python, repository, branch, commit, chamber, and optional-driver status. A missing `pyads`, `serial`, or `windows_capture` produces a visible warning; do not ARM a mode that needs it. Do not create global environment variables ad hoc.

3 Keep O-MBE and Ch-MBE defaults separate

In brief. Every operating default is pending until its chamber owner approves it. Keep the two registers physically and logically separate; never infer from code, screenshots, current selections, or the other chamber.

Every value in both registers is intentionally pending. These entries must not be used to configure, ARM, or troubleshoot either chamber until that chamber's owner supplies and approves them. Until then, follow the applicable chamber SOP and verify the active GUI configuration before ARM. Never copy a value from one chamber to the other.

Chapter 2 gives fixed launcher names and window titles only so the operator can identify the selected software profile. Those identity checks, source-code values, and visible UI selections do not populate either operating-default register.

O-MBE owner-approved defaults - reserved

Configuration item	Owner-approved default
GUI launcher and chamber profile	<i>Pending - to be supplied</i>
RHEED source, mode, source window, ROI, and crop	<i>Pending - to be supplied</i>
Pyrometer mode, port, baud, RTS, and backend	<i>Pending - to be supplied</i>
MISTRAL mode, endpoint, ports, schema, and cell map	<i>Pending - to be supplied</i>
EvapControl mode, window or log source, and schema	<i>Pending - to be supplied</i>
Classifier package, enable state, and shadow state	<i>Pending - to be supplied</i>
Save root, filename prefix, and session naming	<i>Pending - to be supplied</i>
Sampling, continuous capture, and logging intervals	<i>Pending - to be supplied</i>

Approved by / revision / date: *Pending - to be supplied*

Ch-MBE owner-approved defaults - reserved

Configuration item	Owner-approved default
GUI launcher and chamber profile	<i>Pending - to be supplied</i>
RHEED source, mode, source window, ROI, and crop	<i>Pending - to be supplied</i>
Pyrometer mode, port, baud, RTS, and backend	<i>Pending - to be supplied</i>
MISTRAL mode, endpoint, ports, schema, and cell map	<i>Pending - to be supplied</i>
EvapControl mode, window or log source, and schema	<i>Pending - to be supplied</i>
Classifier package, enable state, and shadow state	<i>Pending - to be supplied</i>
Save root, filename prefix, and session naming	<i>Pending - to be supplied</i>
Sampling, continuous capture, and logging intervals	<i>Pending - to be supplied</i>

Approved by / revision / date: *Pending - to be supplied*

A completed value of **Not applicable** is valid only when approved by the corresponding chamber owner; a blank or pending value is not. Neither register replaces its chamber startup SOP.

4 Open the RHEED post-processing labeler

In brief. The labeler is offline. Select one session, ordered prediction/model-spec pairs, and a safe output directory; then build, open, or validate without contacting instruments.

Step 1. Double-click Start RHEED Post-processing Labeler.cmd or its desktop shortcut.

Step 2. Confirm that the application has the **Build** and **Open / Validate** tabs, plus **Build report and open**, **Open report**, **Validate annotations**, and **Open English PDF manual**. It must not start Growth Monitor or contact instruments.

Build a portable report from an archived Growth Monitor session, then review and label saved RHEED frames in the system browser. This tool never connects to instruments or changes the live GUI. Model outputs are visible: exported labels are model-assisted review data and are not eligible as blind-gold training labels.

Build Open / Validate

Session ZIP D:\AI4MBE\scratch\manual-ui-fixture\capture-vqiwjcp\inputs\synthetic_rheed_session.zip Browse...

	Prediction CSV	Model-spec JSON
1	D:\AI4MBE\scratch\manual-ui-fixture\capture-vqiwjcp\inputs\model_1_predictions.csv	D:\AI4MBE\scratch\manual-ui-fixture\capture-vqiwjcp\inputs\model_1_spec.json
2	D:\AI4MBE\scratch\manual-ui-fixture\capture-vqiwjcp\inputs\model_2_predictions.csv	D:\AI4MBE\scratch\manual-ui-fixture\capture-vqiwjcp\inputs\model_2_spec.json

Ordered model pairs

Add pair... Remove Move up Move down

Output directory D:\AI4MBE\scratch\manual-ui-fixture\capture-vqiwjcp\next-report Browse...

Report title Synthetic reconstruction review - operator training

Review WebP quality 78

Build report and open

Ready - generated demonstration inputs

Build and validation messages appear here.

Actual application screenshot. The current desktop labeler Build tab populated with a generated local fixture. The application is offline and no instrument process is running.

Area	Purpose
Session ZIP	Select one archived Growth Monitor session
Model inputs	Pair one prediction CSV with its model-spec JSON on each row
Output directory	Select a new or empty directory, preferably outside Git
Message log	Preserve complete build and validation messages
Open report button	Open a previously generated interactive_report.html
Validate annotations	Check exported JSON against the exact report provenance

This application can run on a computer without instrument software when the archived inputs are complete and the Python environment is valid. Report construction and validation run in an isolated child process so the desktop application remains responsive.

5 Prepare inputs and verify provenance

In brief. The session, predictions, and model specifications must agree exactly on saved-frame order and provenance. Any mismatch stops the build, and exact 1 Hz timing is never assumed.

Session archive

The ZIP archive must contain exactly one member ending in `session_metadata.json`, exactly one member ending in `heartbeat_log.csv`, and every sibling frames/ image referenced by the heartbeat rows. Saved-frame elapsed time, heartbeat index, capture sequence, and timezone-aware capture UTC must each increase strictly; gaps are allowed. Do not extract and rename frames manually.

Read `chamber_id` from the session metadata and confirm that it names the intended O-MBE or Ch-MBE source. When the recorded MISTRAL mode is ADS, also preserve its recorded endpoint, ports, and cell count. Compare provenance only with the corresponding owner-approved register in Chapter 3; never edit an archive to make it resemble the other chamber.

```
growth_session.zip
  session_metadata.json
  heartbeat_log.csv
  frames/
    heartbeat_000001_....bmp
    heartbeat_000002_....bmp
```

Prediction tables and model specifications are positional pairs

File	Required content	Check
Prediction CSV	frame index, heartbeat index, elapsed time, capture UTC, capture sequence, frame name, frame SHA-256, and score columns	Rows match the saved-frame order exactly
Model-spec JSON	schema version 1, non-empty key and title, at least two unique classes, and equally sized unique probability columns	Class order matches the score columns; status and provenance are optional
Session ZIP	metadata, heartbeat log, and frames	Every referenced frame exists in the archive

Prediction `frame_index` may begin at 0 or 1, but it must then remain contiguous. Heartbeat indices, capture sequences, and elapsed times may contain gaps. The tool follows actual saved frames and recorded times; it never assumes exact 1 Hz sampling.

Any mismatch in row count, order, time, sequence, filename, or SHA-256 stops the build. Do not edit the inputs to bypass an error.

Recommended local layout

```
<local-data-root>\2026-08-11_anneal\
  source\growth_session.zip
  predictions\model_A.csv
  specs\model_A.json
  output\
  exports\
```

6 Build and open a report safely

In brief. Build only into a new or empty directory outside Git. Preserve the complete report directory and the original prediction/model-spec inputs separately.

Step 1. Choose the preserved archive copy in Session ZIP.

Step 2. Add one Prediction CSV and Model-spec JSON row for every model to display. Pairing is positional, so preserve row order.

Step 3. Choose a new or empty directory outside the repository. The desktop labeler will not overwrite a non-empty directory.

Step 4. Enter a report title. Review quality changes only the lossy review images; it never changes archived pixels or model predictions. The default value of 78 is suitable for routine use.

Step 5. Click **Build report and open**. Wait for success and for `interactive_report.html` to open automatically.

PowerShell fallback

```
Set-Location '<GUI repository>'
conda activate ai4mbe-gui
python -m tools.rheed_postprocessing_labeling build `
  --session '<local-data-root>\growth_session.zip' `
  --predictions '<local-data-root>\predictions.csv' `
  --model-spec '<local-data-root>\model_spec.json' `
  --output-dir '<local-data-root>\labeling-output'
```

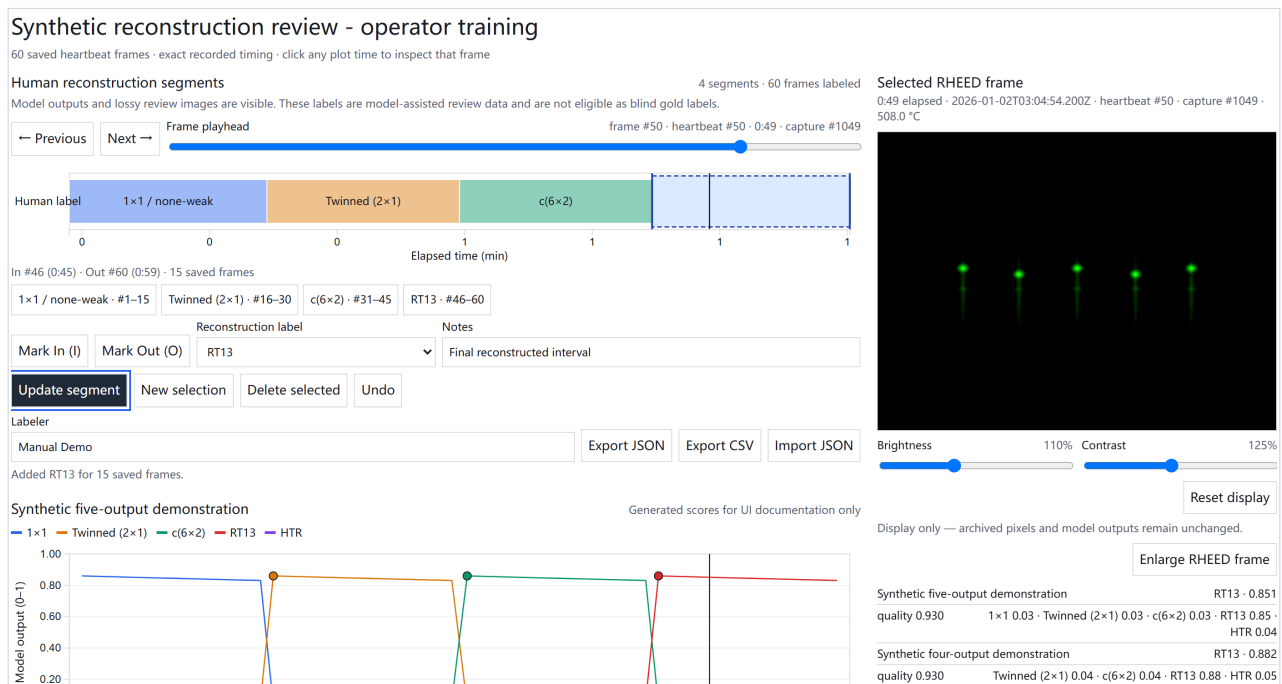
Copy the entire report directory

The report consists of `interactive_report.html`, `images/`, `vendor/`, and `run_manifest.json`. Copy the entire directory during handoff. Preserve the original prediction CSV and model-spec JSON separately outside the report directory.

The desktop application always refuses a non-empty output directory. CLI `--overwrite` may replace only a recognized, unmodified tool output. It validates the new inputs and stages a complete replacement first. If the old report changes during the build, replacement stops and the original content is restored. Filesystem roots, repository paths, the current directory, and directories containing inputs are always rejected.

7 Label temporal segments like an editing timeline

In brief. Mark inclusive In and Out frames, choose the human label, and save exact endpoint provenance. Adjacent segments are allowed; overlaps are rejected.



Actual application screenshot. The current English browser editor with generated RHEED frames, synthetic model curves, saved human segments, and a synchronized playhead.

Step 1. Move the Frame playhead or click a model plot to locate the first saved frame of a segment. Click **Mark In**.

Step 2. Move to the final saved frame of the segment and click **Mark Out**. The Out marker is inclusive.

Step 3. Choose a reconstruction label, enter the labeler name, add notes when useful, and click **Add segment**.

Step 4. Select a saved segment to edit it. Update preserves its annotation ID. Adjacent segments are allowed; overlapping segments are rejected.

The displayed and exported saved-frame ordinal is 1-based. Every endpoint also records heartbeat index, capture sequence, UTC, and frame SHA-256. This preserves traceability even when the sampling interval or sequence contains gaps.

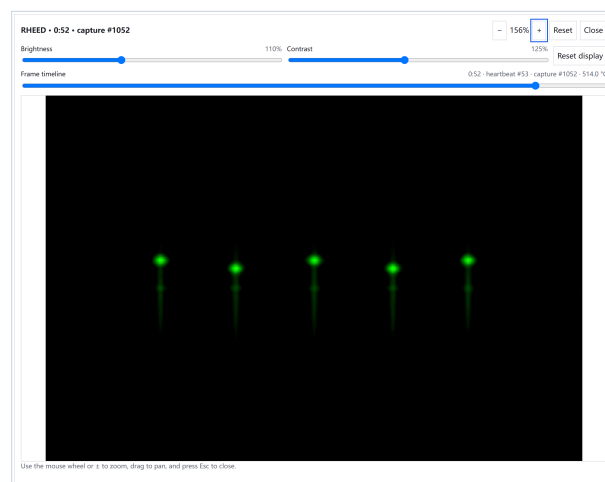
Labels cover the saved frames from In through Out, inclusive. They do not claim that unsaved camera frames or every capture-sequence number in between was reviewed.

8 Review, display controls, and drafts

In brief. Keep the image, playhead, model guides, and enlarged timeline synchronized. Display adjustments never change source data, and browser drafts do not replace frequent JSON exports.

All time indicators remain synchronized

Action	Must update together	Must not change
Move the main timeline	Current RHEED image, model guides, enlarged-view timeline	Archive and prediction values
Click a model curve	Main playhead, image, and metadata	Saved segments
Adjust brightness or contrast	On-screen appearance	Pixels, SHA-256, and model outputs
Enlarge the image	Zoom view and its adjustable timeline	Source image and segment boundaries



Actual application screenshot. The actual enlarged-frame dialog. Its timeline changes the selected frame while preserving zoom, pan, and display adjustments.

Drafts and editing

- The browser stores a draft scoped by both dataset ID and model-context fingerprint in localStorage. This is not a durable backup.
- Export JSON frequently, especially during long reviews and before changing computers, browsers, or cache settings.
- Delete selected requires confirmation. Undo restores only the most recent annotation mutation.
- Import JSON validates before replacement. On failure, the current segment set remains unchanged.

Interpretation

Labels describe the dominant surface-reconstruction category seen by the reviewer during the interval. The displayed **Uncertain** label exports as unknown; **1x1 / none-weak** exports as none_weak. Never overwrite a human selection with model argmax, and never interpret unknown as acquisition-quality rejection.

Brightness, contrast, zoom, and image enlargement aid inspection only. Brightness and contrast are stored with each segment; zoom is not exported. The report never rewrites archived images or model outputs.

Use a four-pass review rhythm: survey the full run, confirm every In and Out frame, inspect gaps and uncertain labels, then export and validate the canonical JSON.

9 Export, import, and fail-closed validation

In brief. JSON is the canonical annotation artifact. Validate it against the exact report and stop on any run, model-context, endpoint, label, or overlap mismatch.

JSON is the canonical artifact

Use Export JSON to resume work, validate provenance, and support downstream processing. Export CSV is useful for meetings and tabular analysis, but it does not replace the complete nested provenance in JSON.

Binding	Purpose
dataset_id and source archive SHA-256	Bind the source session
Ordered-frame fingerprint	Bind frame order and per-frame SHA-256
model_context_fingerprint	Bind the visible predictions and model specifications
Endpoint provenance	Bind saved-frame ordinal, heartbeat, UTC, sequence, and SHA-256
model_outputs_visible=true	Disclose that the models were visible
eligible_for_gold=false	Prevent blind-gold misuse

Validate in the desktop application

Step 1. Select the original `interactive_report.html`.

Step 2. Select the exported annotation JSON.

Step 3. Click **Validate annotations**. Continue only when the application says **Annotation JSON is valid for this report**.

PowerShell validation

```
python -m tools.rheed_postprocessing_labeling validate `
  --report '<local-data-root>\labeling-output\interactive_report.html' `
  --annotations '<local-data-root>\exports\segment_annotations.json'
```

A wrong run, changed model context, altered endpoint, overlapping segment, invalid label, or gold-data claim causes validation to fail. Return to the correct report and original export and investigate the exact message.

Browser Import performs immediate client-side provenance checks before replacing the current draft. Desktop **Validate annotations** is the authoritative full fail-closed validation.

10 Data safety and scientific boundaries

In brief. Keep laboratory data, checkpoints, reports, and annotations out of Git. These labels are model-assisted, not blind gold, and reconstruction is distinct from QC and film quality.

What belongs in Git

May be tracked	Keep out by default
Tool code, templates, tests, and documentation	Real session ZIP files, raw RHEED images, and sensor logs
Example model specifications	Real predictions, generated reports, and annotation JSON or CSV
Sanitized documentation screenshots made with generated demo inputs	Checkpoints, unpublished results, and experiment-specific screenshots

Safe handling sequence

Step 1. Keep the source session ZIP read-only or preserve an immutable copy, and record its SHA-256.

Step 2. Create a dedicated working directory outside the repository. Keep reports, exports, and screenshots there.

Step 3. For handoff, copy the complete report directory and canonical JSON. Separately preserve the original prediction CSV, model-spec JSON, and SHA manifest. Never send only the HTML.

Step 4. Run statistics or training-data preparation only after validation, and preserve the original export unchanged.

Model plots and lossy review images are visible during labeling. These annotations are eligible only as model-assisted review. Formal blind-gold labels require a separate workflow that hides model and Equalizer outputs and preserves the required audit evidence.

The current classification target is bare STO surface reconstruction before growth. Do not interpret a reconstruction label as FeSe film quality. Do not equate reconstruction unknown with acquisition-quality QC_REJECT.

11 Troubleshooting

In brief. Preserve evidence, then use the branch for the chamber named in the exact window title and launcher log. Never repair one chamber by copying settings from the other.

First preserve the exact error text, occurrence time, branch, commit, and selected paths. Do not begin by deleting environments, changing global variables, or force-resetting Git. A live-driver warning is fail-visible behavior, not a prompt for ad hoc dependency installation.

O-MBE live-GUI branch

Use `Start O-MBE Growth Monitor.cmd`; the exact title is **Oxide MBE Growth Monitor**. If the title differs, close normally, preserve the latest launcher log, and restart only with the O-MBE launcher. Compare selected interfaces only with the owner-approved O-MBE register in Chapter 3; a pending entry is a stop condition, not a reason to borrow a Ch-MBE value.

Ch-MBE live-GUI branch

Use `Start Ch-MBE Growth Monitor.cmd`; the exact title is **Chalcogenide MBE Growth Monitor**. If the title differs, close normally, preserve the latest launcher log, and restart only with the Ch-MBE launcher. Compare selected interfaces only with the owner-approved Ch-MBE register in Chapter 3; a pending entry is a stop condition, not a reason to borrow an O-MBE value.

Shared live and offline symptoms

Symptom	Likely cause	Safe action
Desktop shortcut is absent or opens an old checkout	Installer was not run after the checkout moved	Run the shortcut installer from the current repository root
Growth Monitor title names the wrong chamber	Wrong launcher or stale shortcut	Close normally; follow the matching chamber branch above
Launcher window closes immediately	Environment, dependency, or pre-flight failure	Read the newest matching-chamber launcher log; run the same CMD from PowerShell only to preserve its error
Optional-driver warning appears	A selected live mode may lack its Python driver	Record the missing module; do not change mode unless the chamber owner has approved the alternative
Temperature, RHEED, MIS-TRAL, or EvapControl is invalid	Interface, vendor window, selected mode, or stale source issue	Record chamber, mode, connected/valid/error, sequence, and age; do not guess settings
Labeler rejects Build	Missing input, mismatched pair, or non-empty output	Check the session, each positional pair, and a new output directory
Provenance mismatch	Predictions do not belong to the ZIP or rows changed	Find the matching predictions and spec; do not edit the CSV
HTML has no images or curves	Only the HTML was copied or assets are missing	Restore the complete report directory and relative paths
Draft disappeared	Browser, computer, or localStorage changed	Import the latest JSON and export more frequently
Import or validation fails	Run, context, endpoint, overlap, or label mismatch	Use the original report and JSON; read the fail-closed message

Minimum evidence for the maintainer

- Full error text and occurrence time. Do not report only that it does not work.
- Output of `git status --short --branch` and `git rev-parse HEAD`.
- Chamber, exact window title, launcher name, launcher-log path, environment path, and selected modes. For labeling issues, add report-manifest and annotation-JSON SHA values; raw images are not initially required. Keep sensitive data out of public channels.

12 Version verification and quick checklist

In brief. Record versions and hashes, complete only the startup checklist for the selected chamber, then use the shared offline checklist. Stop whenever identity, instrument state, provenance, or validation cannot be explained.

Record versions before acquisition or labeling

```
Set-Location '<GUI repository>'
git status --short --branch
git rev-parse HEAD
conda env list
conda activate ai4mbe-gui
python --version
python -m tools.rheed_postprocessing_labeling --help
```

Use the resolved Python interpreter and repository path recorded by the matching chamber launcher or labeler launcher. Read AI4MBE_GUI_PYTHON only when it is already configured on that workstation; do not create it ad hoc. If the branch or commit differs from the team-specified version, or Git reports unknown modifications, stop and verify.

O-MBE startup checklist

- ☐ Use Start O-MBE Growth Monitor.cmd; never use the Ch-MBE launcher for this chamber.
- ☐ Confirm the exact title **Oxide MBE Growth Monitor**.
- ☐ Confirm the launcher log identifies O-MBE, the intended repository, branch, commit, Python, and required optional drivers.
- ☐ Compare every selected interface and logging field only with the approved O-MBE register in Chapter 3; stop on any pending or unexplained value.
- ☐ Confirm advancing RHEED sequence, valid temperature and instrument states, data age, and intended log directory under the O-MBE SOP.
- ☐ ARM only when the authorized operator and O-MBE SOP permit it; this checklist grants no operating or setpoint authority.

Ch-MBE startup checklist

- ☐ Use Start Ch-MBE Growth Monitor.cmd; never use the O-MBE launcher for this chamber.
- ☐ Confirm the exact title **Chalcogenide MBE Growth Monitor**.
- ☐ Confirm the launcher log identifies Ch-MBE, the intended repository, branch, commit, Python, and required optional drivers.
- ☐ Compare every selected interface and logging field only with the approved Ch-MBE register in Chapter 3; stop on any pending or unexplained value.
- ☐ Confirm advancing RHEED sequence, valid temperature and instrument states, data age, and intended log directory under the Ch-MBE SOP.
- ☐ ARM only when the authorized operator and Ch-MBE SOP permit it; this checklist grants no operating or setpoint authority.

Integrity hashes

```
Get-FileHash '<local-data-root>\growth_session.zip' -Algorithm SHA256
Get-FileHash '<local-data-root>\exports\segment_annotations.json' -Algorithm SHA256
Get-FileHash '.\docs\RHEED_GUI_Postprocessing_Labeling_User_Manual_EN.pdf' -Algorithm SHA256
```

Companion text and AI prompts

Searchable manual: docs/RHEED_GUI_Postprocessing_Labeling_User_Manual_EN.md

AI prompt pack: docs/RHEED_GUI_Postprocessing_Labeling_AI_Prompt_Pack_EN.md

The prompt pack does not authorize an AI to invent pending defaults or make instrument-control decisions.

Shared offline and handoff checklist

- ☐ Preserve a read-only source ZIP and its SHA-256.
- ☐ Confirm session metadata names the intended chamber; never repair provenance by editing the archive.
- ☐ Open the offline labeler with the dedicated launcher.
- ☐ Pair every prediction CSV with the correct model-spec JSON.
- ☐ Use a new or empty output directory outside Git.
- ☐ Review boundaries frame by frame; allow no overlaps.
- ☐ Export JSON and validate it against the original report.
- ☐ Record branch, commit, environment, manifest, and export hashes.
- ☐ Handoff the complete report and declare model-assisted, not blind-gold.

Stop and contact the maintainer when instrument state, stale data, version identity, provenance, or validation cannot be explained.